

CNSCC.160 – Fundamentals of Communication

Systems Tutorial Questions and Revision (Satellite Communications)

KNOWLEDGE

1. Define the terms **longitude**, **latitude**, **azimuth** angle and **elevation** angle.
2. What does the acronym '**DBS**' stand for and what is a DBS service?
3. A satellite revolves around the Earth in an **elliptical** orbit. When does the satellite reach its **maximum speed** and when does it reach its **minimum speed**?
4. Define the terms **geosynchronous** and **geostationary orbit**.
5. How is the **gain of an antenna** affected when (i) its beam-width increases, (ii) its diameter increases, (iii) its operating frequency reduces?

6. The **Inmarsat** system uses 4 geostationary satellites to provide its mobile satellite service. The competing **Iridium** system uses a network of 66 satellites in low Earth orbit instead. List three advantages of the Inmarsat system and three advantages of the Iridium system.

UNDERSTANDING

1. In a satellite TV link (e.g., from the uplink station to your home via a satellite) the antenna at the uplink station is much bigger than the antenna on your home. Why are they not both the same size?

2. Without making any adjustments, do you think it is possible that a standard fixed (non-steerable) domestic satellite dish can receive and recover signals from more than one satellite?